

Deep Learning — Tutorial 2

Numerical Computation: Goodfellow, Bengio & Courville, Chapter 4

Practice set — not for submission

These questions are for your own practice and do not need to be submitted or graded. Work them through properly and write out each derivation in full rather than reading the question and deciding you could probably do it. The gap between those two states is where most of the learning happens.

Q1. Overflow and underflow

- (1) Compute $\text{softmax}(\mathbf{x})$ for $\mathbf{x} = (0, 0, 0)$ by hand. What probability distribution do you get, and why?
- (2) For $\mathbf{x} = (3, 1, 2)$, compute $\text{softmax}(\mathbf{x})$ to 3 decimal places
 - (a) directly, and
 - (b) using the stabilisation trick (subtract $c = \max_i x_i$).
 Confirm the two results are identical.
- (3) Consider $\mathbf{x} = (1000, 1001, 1002)$.
 - (a) Which quantity in the naive softmax computation overflows?
 - (b) Write the shifted vector $\mathbf{x} - c$ used by the stable version.
 - (c) Now consider $\mathbf{x} = (-1000, -1001, -1002)$: which failure mode (overflow or underflow) occurs naively, and how does the shift fix it?
- (4) Prove that softmax is invariant to a constant shift, i.e. for any scalar c , $\text{softmax}(\mathbf{x}) = \text{softmax}(\mathbf{x} - c)$. Then explain, using your result, why computing $\log \text{softmax}(\mathbf{x})_i$ naively can still underflow, and write down a numerically stable formula for $\log \text{softmax}(\mathbf{x})_i$.

Q2. Poor conditioning

- (1) A symmetric matrix has eigenvalues $\lambda_1 = 20$ and $\lambda_2 = 0.4$. State its condition number κ .
- (2) For $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 0 & 0.03 \end{pmatrix}$, write down its eigenvalues and condition number. A perfectly conditioned matrix has $\kappa = 1$; roughly how many decimal digits of accuracy (i.e. $\log_{10} \kappa$) can a linear solve with this $\mathbf{A}$ lose in the worst case? Would you therefore call it well- or ill-conditioned relative to $\kappa = 1$?
- (3) Let $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.
 - (a) Find its eigenvalues and condition number.
 - (b) Solve $\mathbf{A}\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = (3, 3)$ and again for $\mathbf{b} = (3, 3.3)$.
 - (c) Compute the *relative* error in the output, $\|\Delta \mathbf{x}\| / \|\mathbf{x}\|$, and the *relative* perturbation in the input, $\|\Delta \mathbf{b}\| / \|\mathbf{b}\|$. Form their ratio and check that it does not exceed the condition number κ from part (a).

Q3. Gradient-based optimization

- (1) For $f(x) = x^2$ with $x_0 = 3$ and learning rate $\varepsilon = 0.1$, write the update as $x_{t+1} = r x_t$. Find r and compute x_1, x_2, x_3 .
- (2) Let $f(x, y) = x^2 + 4y^2$. Compute ∇f at the point $(1, 1)$ and perform one step of gradient descent with $\varepsilon = 0.1$. What is the new point?
- (3) For $f(x) = x^2$ the gradient-descent update factor is $(1 - 2\varepsilon)$.
 - (a) For which range of ε does the iteration converge to 0?
 - (b) Which value of ε makes it oscillate forever without converging?
 - (c) What happens for $\varepsilon = 1.1$?
- (4) Let $f(x, y) = x^2 + xy + y^2$. Compute ∇f at $(1, 2)$, and give the *unit* direction of steepest descent at that point.

Q4. Curvature, the Hessian, and Newton's method

- (1) Let $f(x, y) = x^2 + y^2 - 4x$. Find the critical point, write the Hessian $\mathbf{H}$, and classify the critical point using the second-derivative test.

- (2) Let $f(x, y) = x^2 - 3y^2$. Find the critical point and its Hessian, and show that it is a saddle point.
- (3) Apply one step of Newton's method to $f(x) = x^2 - 6x + 5$ starting from $x_0 = 0$. Verify that you land exactly on the minimiser, and explain why a single step suffices.
- (4) Consider $f(x) = x^4$, whose minimum is at $x^* = 0$.
 - (a) Show Newton's update is $x \leftarrow \frac{2}{3}x$.
 - (b) Starting from $x_0 = 3$, compute x_1, x_2, x_3 .
 - (c) Convergence here is only *linear*, not the usual fast quadratic convergence of Newton's method. Explain why, referring to the curvature $f''(x)$ at the minimum.
- (5) Let $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \mathbf{H}\mathbf{x}$ with $\mathbf{H} = \text{diag}(1, 10)$. At $\mathbf{x} = (1, 1)$ the gradient is $\mathbf{g} = (1, 10)$. Compute the optimal step size for a step along $-\mathbf{g}$,

$$\varepsilon^* = \frac{\mathbf{g}^\top \mathbf{g}}{\mathbf{g}^\top \mathbf{H} \mathbf{g}}.$$

Compare ε^* with $1/\lambda_{\max}$ and $1/\lambda_{\min}$ of $\mathbf{H}$, and explain why the largest eigenvalue forces ε^* to be small even though the $\lambda = 1$ direction could tolerate a much larger step. What does this mismatch between the two eigenvalues (the conditioning of $\mathbf{H}$) imply for the speed of gradient descent?

Q5. Constrained optimization (KKT / Lagrange)

- (1) Using a Lagrange multiplier, minimise $f(x, y) = x^2 + y^2$ subject to the equality constraint $x + 2y = 5$. Report x^* , y^* , λ , and $f_{\min}$.
- (2) Minimise $f(x, y) = (x - 3)^2 + (y - 3)^2$ subject to the inequality constraint $x + y \leq 4$. Using the KKT conditions:
 - (a) write the generalised Lagrangian;
 - (b) determine whether the constraint is *active* (use complementary slackness to justify);
 - (c) solve for x^*, y^*, α and $f_{\min}$.
- (3) State the four KKT conditions in general form for minimising $f(\mathbf{x})$ subject to $h(\mathbf{x}) \leq 0$, and explain in one sentence what *complementary slackness* ($\alpha h(\mathbf{x}) = 0$) means intuitively.

Q6. Linear least squares

- (1) Fit a line $y = c_0 + c_1x$ to the data points $(0, 1), (1, 3), (2, 4)$ by solving the normal equations $\mathbf{A}^\top \mathbf{A} \mathbf{c} = \mathbf{A}^\top \mathbf{b}$. Give the fitted line.
- (2) Starting from $f(\mathbf{x}) = \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2$:
 - (a) derive ∇f and the Hessian $\mathbf{H}$;
 - (b) show $\mathbf{H} = \mathbf{A}^\top \mathbf{A}$ is positive semidefinite, hence f is convex;
 - (c) conclude that any solution of the normal equations is a *global* minimiser.

Q7. Short conceptual answers

Answer each in **two or three sentences only**.

- (a) Distinguish between a problem that is hard because of a poor *algorithm* and one that is intrinsically *ill-conditioned*. Why can no choice of algorithm rescue an ill-conditioned problem in finite-precision arithmetic, and how does the condition number quantify this?
- (b) Newton's method reaches the minimum of a quadratic in a single step, yet deep networks are trained almost entirely with (stochastic) gradient descent. Give *two* reasons second-order methods are impractical for large neural networks.
- (c) Explain the link between *poor conditioning* (Q2) and *gradient-based optimization* (Q3): how does the condition number of the Hessian affect the behaviour of gradient descent, and what does this look like geometrically (think of the level sets)?
- (d) The softmax stabilisation trick works because of a specific *invariance* of the function. State that invariance, and articulate the *general lesson* it teaches about implementing mathematical expressions numerically.